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Text- and game-based communication for climate change attitude, self-efficacy, and behavior: A controlled experiment

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ABSTRACT

The advantages of both gamification and immersive technologies are often advocated for in contexts in which human motivation is lacking. This is the case with environmental sustainability and climate change engagement, where knowledge alone is not sufficient for pro-environmental behavior. However, the existing literature lacks rigorous studies comparing games and immersive media to more traditional communication methods. In this article, we describe an experiment (N=105) where participants used a climate change game in PC or immersive VR or were assigned to a text-based control. Our findings suggest that all three conditions increased climate change attitudes and environmental self-efficacy, but no significant differences were found between them. Although VR players tended to enjoy their experience significantly more than other participants, we did not find significant differences in self-reported immersion. Furthermore, neither enjoyment nor immersion correlated with attitude or self-efficacy shifts. Our exploration of participant behavior yielded similar results for all three conditions, both in commitment to action and self-reported completion ten days after the intervention. Our results suggest that games can improve attitudes and self-efficacy even in highly involved audiences, but their comparative advantage in certain contexts may be smaller than commonly assumed.

1. Introduction

As the climate crisis seriously threatens ecologic, economic, societal and cultural systems, urgent action towards reducing greenhouse gas emissions and coping with its impacts in the near-term is fundamental (Intergovernmental Panel on Climate Change, 2022). Over two decades of research on climate change engagement have yielded numerous insights on how people connect with the issue (Kollmuss & Agyeman, 2002; Lorenzoni, Nicholson-Cole, & Whitmarsh, 2007; Whitmarsh, Lorenzoni, & O'Neill, 2012) and multiple suggestions on how to increase engagement and support those who are already engaged (Wibeck, 2014). These include interactive visualizations (Moser, 2010; Sheppard, 2012; Wibeck, Neset, & Linnér, 2013) and, more broadly, experiential environments where people can make sense of the issue and their own position towards it Monroe, Plate, Oxarart, Bowers, and Chaves (2019).

Gamification, or the transformation of activities to afford game experiences (Hamari, 2019), is one of such proposed methods, and has been generally successful in promoting cognitive, affective, and behavioral engagement with climate change (Fernández Galeote et al., 2021). Given that gamification often refers to the application of game design elements in non-game contexts (Deterding, Dixon, Khaled, & Nacke, 2011) as opposed to full-fledged serious games, we clarify that in this study it is used as an umbrella term for any form of gameful engagement that seeks outcomes beyond entertainment, including serious games and game-based learning (Hamari, 2019). This use of the word has been adopted in recent climate change engagement literature (Douglas & Brauer, 2021; Fernández Galeote et al., 2021; Rajanen & Rajanen, 2019) and beyond (Koivisto & Hamari, 2019; Krath, Schürmann, & Von Korflesch, 2021) and points towards the importance of exploring different artifact configurations as part of an overarching phenomenon.

When applying gamification, motivation to engage with contents and enact behaviors (Plass, Homer, & Kinzer, 2015; Ryan, Rigby, & Przybylski, 2006) is typically sought through virtual game worlds that encourage direct experience and exploration as memorable ways of engaging with a topic (Monroe et al., 2019; Plass et al., 2015) as well as stories, lived events and characters that tend to elicit enjoyment and emotional engagement (Hemenover & Bowman, 2018; Plass et al., 2015). Another promising avenue is immersive virtual reality (VR), which is not only typically engaging (Pellas, Dengel, & Christopoulos, 2020) but also allows users to feel a sense of presence, or being in the virtual environment (Cummings & Bailenson, 2016). In this way, climate change can become immediate and concrete rather than remote and intangible (Breves & Schramm, 2021; Markowitz, Laha, Perone,

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Pea, & Bailenson, 2018), and thus more emotionally salient (Markowitz & Bailenson, 2021).

However, the existing literature on gamified and game-based climate change engagement lacks various important research avenues, including the use of potentially effective communication frames, the examination of immersive VR games, a focus on engagement aspects besides knowledge, and the use of rigorous experimental designs comparing the effects of games to informationally equivalent conditions, for example texts (Fernández Galeote et al., 2021). Similarly, the literature on immersive environmental persuasion typically focuses on non-interactive conditions, such as 360-degree videos, and lacks empirical evidence of effectiveness (Breves & Greussing, 2021). Therefore, and despite the generally positive results associated with these media and technologies (Breves & Greussing, 2021; Fernández Galeote et al., 2021), our knowledge remains fragmented and its empirical support limited, especially when comparing games and immersive technologies to other forms of communication.

Contributing to address the aforementioned gaps, this article describes an experiment (N = 105) involving a narrative climate change game that frames climate change as a wellbeing issue and involves its anthropic causes, mechanisms, and consequences for non-human and human life, especially infectious diseases and pandemics. Participants either played the game on a computer screen or immersive VR, or were assigned to a text-based informational equivalent acting as a control condition. Our study examines the effects of playing the game or reading the text on climate change attitude, environmental self-efficacy, and pro-environmental intentions (PEI) and behaviors (PEB). Our findings suggest that games on traditional screens and immersive VR can be as effective as text-based communication methods in promoting these aspects of engagement, thus complementing previous findings involving similar comparisons using games (Soekarjo, van Oostendorp, et al., 2015) and extending them into the area of interactive immersive VR.

2. Background

2.1. Enjoyment as part of the game experience

This study concerns gamification understood as the result of transforming an activity to afford a game-like experience with the intention to bring a change, be it cognitive or behavioral, through discrete game elements or full-fledged games (Hamari, 2019). Game experiences typically include outcomes such as enjoyment, which can be seen as the satisfactory feeling associated with a positive experience (Cairns, Cox, & Nordin, 2014).

Thus, the first two hypotheses refer to the potential of game-based communication in VR and PC to elicit interest and enjoyment. Educational games have been found to be more engaging than comparable texts (Arici, 2008) and other learning methods (Lieberoth, 2015), in some cases with large effect sizes (McLaren, Adams, Mayer, & Forlizzi, 2017). Gamification is supported by theories that assume positive states such as flow and intrinsic motivation as part of the core experience of playing (Krath et al., 2021). This leads us to hypothesize:

- H1.1: Playing the game in any form will be significantly more enjoyable than reading the text.

In addition, immersive VR has been perceived as more enjoyable than screen-based media (Makransky, Andreasen, Baceviciute, & Mayer, 2021; Makransky & Mayer, 2022; Pellas et al., 2020; Reer, Wehden, Janzik, Tang, & Quandt, 2022). This is not surprising since new and complex experiences tend to elicit curiosity (Berlyne, 1954). Similarly, a "novelty effect" has been reported in gamification (Hamari, Koivisto, & Sarsa, 2014). Therefore, we hypothesize:

- H1.2: Playing the game in VR will be more significantly enjoyable than playing on PC.

2.2. Immersion as part of the game experience

Immersion is also typically seen as part of the game experience. This is a concept used in game studies to refer to the emotional and attention involvement experienced when doing an activity (Cairns et al., 2014) and which can occur via challenges, imagination (which would make a text potentially immersive), and sensory aspects (Ermi & Mäyrä, 2007).

Our next hypothesis refers to the immersive qualities of the treatment conditions. This study focuses on immersion as a construct that signals involvement with the game rather than spatial presence, or the feeling of being transported to a different space (Cairns et al., 2014). However, presence has similarities with sensory immersion (Ermi & Mäyrä, 2007) and the technological immersiveness of a medium typically has a considerable effect on the sense of presence (Cummings & Bailenson, 2016). Technological immersiveness refers to a technology's capacity to present the user with a vivid virtual environment while isolating them from their surroundings; thus, technologies capable of offering multisensory and high fidelity simulations, a correspondence between physical and virtual body action, and narratives that transport the user away from their physical surroundings are more immersive than those not including these aspects (Slater & Wilbur, 1997). Consequently, a simulation in immersive VR is more technologically immersive than a simulation on a PC screen, which is more immersive than a text-based description of the simulation. In addition, the evidence reported by meta-reviews (Cummings & Bailenson, 2016) and individual studies tends to show higher presence in VR than in 2D screen (Shu, Huang, Chang, & Chen, 2019; Wagler & Hanus, 2018), although it must be noted that presence has a personal component, or immersive tendency, which may be affected by other psychological variables (Weibel, Wissmath, & Mast, 2010).

In addition to sensory immersion, games may also increase immersion by providing challenges, which our text-based control does not include (Ermi & Mäyrä, 2007). Therefore, it is hypothesized that participants will report higher levels of immersion in the game conditions, and in VR in particular due to its higher potential for cognitive absorption (Kampling, 2018), even comparable to that of real-world activities (Lui & Goel, 2022). Thus, we hypothesize:

- H2: Technological immersiveness will be positively associated with self-reported immersion.

2.3. Climate change attitude

Climate change engagement involves three forms of connection to the issue: cognitive (knowing), affective (caring), and behavioral (acting) (Lorenzoni et al., 2007). Rather than being limited to awareness and knowledge, engagement considers the internal and external factors, from attitudes to economic constraints and social norms, that influence a person's relationship to climate change (Gifford & Nilsson, 2014; Lorenzoni et al., 2007; Whitmarsh et al., 2012).

During the last two decades, the complexity of climate change engagement has been explored leading to significant advances in its scientific understanding and promotion (see, e.g., Gifford & Nilsson, 2014; Kollmuss & Agyeman, 2002; Lertzman, 2019; Lorenzoni et al., 2007; Whitmarsh et al., 2012). Suggested support methods include the use of specific message frames (Badullovich, Grant, & Colvin, 2020), digital visualizations (Moser, 2010; Sheppard, 2012; Wibeck et al., 2013), interactive environments (Monroe et al., 2019; Sterman, 2011; Wibeck, 2014), and dialogic processes (Moser, 2010; Wibeck, 2014), all of which point to the potential of gamification as an experiential method.

Multiple variables have been included in pro-environmental behavior models, including awareness of consequences, responsibility, personal norms (Schwartz, 1977), values, beliefs in threats and possibility for restorative action (Stern, Dietz, Abel, Guagnano, & Kalof, 1999), personality traits, social and cultural factors, and other conditions (Kollmuss & Agyeman, 2002). This study examines climate

change attitude as an important precedent to PEI and PEB (Ajzen, 1991; Casaló & Escario, 2018; Klöckner, 2013). Attitude can be defined as a combination of beliefs, affect, and intentions towards an activity or issue (Schultz et al., 2005).

In this vein, our third set of hypotheses considers the role of the intervention on climate change attitude. A meta-review has shown that interventions targeting climate change attitudes amount to a small positive effect irrespective of the type of intervention (Rode et al., 2021). Indeed, persuasive texts can have an attitudinal effect on climate change (Sinatra, Kardash, Taasoobshirazi, & Lombardi, 2012), as can game-based interventions in various formats (Fernández Galeote et al., 2021). Therefore, we hypothesize:

- H3.1: The intervention in any of its forms will positively affect participants' climate change attitude.

The existing theoretical bases support the notion that games and gamification may lead to heightened attitudinal effects when compared to less immersive or interactive media (Janakiraman, Watson, & Watson, 2018), as could immersive VR (Breves & Greussing, 2021). However, a dearth of comparisons between games and more traditional media in the area of climate change engagement has been observed (Fernández Galeote et al., 2021), and in particular in respect to attitudes (Soekarjo et al., 2015). A similar issue exists in the lack of robust empirical designs in gamified sustainability education research (Hallinger, Wang, Chatpinyakoop, Nguyen, & Nguyen, 2020) and in regards to empirical evidence of VR's effectiveness in shifting environmental attitudes (Breves & Greussing, 2021). When comparing a computer game about climate change to a control condition involving the same information in text and image, no significant differences were found (Soekarjo et al., 2015). Similarly, a comparison of VR- and non-VR-based conditions depicting the effects of climate change on a glacier saw an increase in awareness across conditions, but found no differences between the two (Thoma et al., 2023).

Despite these shortcomings, studies in other areas have found that both screen and immersive VR stimuli impacted attitudes (Bujić, Salminen, Macey, & Hamari, 2020) and immersive VR videos were more effective than regular screen ones (Breves & Heber, 2020; Filter, Eckes, Fiebelkorn, & Büssing, 2020; Fonseca & Kraus, 2016). Technological advances such as immersive VR have been key in the emotional turn in journalism, which has clear attitudinal implications as it refers to new forms of emotion-driven reporting seeking to create closer and trusting relationships with audiences (Lecheler, 2020; Sánchez Laws, 2020). Closer to climate change, the study comparing depictions of climate change effects on a glacier found a significant increase in environmental awareness for the VR but not for the non-VR conditions, despite the lack of significant difference between the two (Thoma et al., 2023). In another study, VR embodiment of animals resulted in larger connectedness to nature than watching a video (Ahn et al., 2016). In other cases, the level of interactivity has been linked to shifts in attitudes (Sundar, Kalyanaraman, & Brown, 2003). It has also been found that games can raise more interest in climate change issues than a similar website (Nussbaum et al., 2015). We hypothesize:

- H3.2: The game-based conditions will result in a larger climate change attitude shift than reading the text.
- H3.3: Technological immersiveness will be positively associated with a shift in climate change attitude.

In relation to the above, self-reported immersion could be expected to relate to climate change attitude. Previous studies have observed correlations between screen-based and immersive VR video and human rights attitudes (Bujić et al., 2020) or theorized that narrative immersion can lead to belief in content and attitudinal effects (Green, 2021). In terms of environmental attitudes, higher reported levels of presence have been linked to more concern for the environment (Markowitz et al., 2018) and environmental awareness (Thoma et al., 2023), and vividness of stimulus has been linked to pro-environmental behavior (Bailey et al., 2015). More broadly, studies on the effect of VR on attitudes and behavior report generally positive findings, which is commonly explained theoretically through spatial presence and embodiment (Breves & Greussing, 2021). However, these studies either focus on different areas of study or involve designs that are quite different to ours, with considerable importance of player avatars. We also hypothesize:

- H3.4: Self-reported immersion will be positively associated with a shift in climate change attitude.

2.4. Environmental self-efficacy

Environmental self-efficacy is another important antecedent to PEI and PEB (Ajzen, 1991; Casaló & Escario, 2018; Klöckner, 2013). Self-efficacy, which refers to a person's confidence in their capacity to overcome difficulties in acting (Moeller & Stahlmann, 2019), is studied in our final set of hypotheses. The proposed intervention is not a simulation of behavior where people directly practice detailed PEBs. Rather, it is mostly based on simple actions, verbal argumentation and suggestive imagery. However, apart from mastery experiences and vicarious learning, people can develop self-efficacy through social persuasion and physiological and affective states (Lehikko, 2021). By providing support against amotivation (Pelletier, Dion, Tuson, & Green-Demers, 1999) and showing actions that can help mitigate climate change, all three conditions attempt to promote both information acquisition and emotional states conducive to action, from concern to anticipated positivity. As interventions based on knowledge transmission have increased efficacy beliefs (Geiger, Swim, & Fraser, 2017), and previous interventions have shown that both immersive VR and text can increase people's beliefs in their own capacity to affect the environment (Ahn, Bailenson, & Park, 2014), we hypothesize:

- H4.1: The intervention in any of its forms will positively affect participants' environmental self-efficacy.

Further, and despite the aforementioned lack of comparisons between games and equivalent informational content, as well as a lack of research of game-based climate change engagement in VR (Fernández Galeote et al., 2021), studies involving PC and VR-based educational training tend to report heightened self-efficacy (Lehikko, 2021). In stimuli related to environmental topics, immersive virtual environments have been found to improve internal environmental locus of control compared to video and text, an effect that may be attributed to perceptual richness and interaction (Ahn et al., 2014). Training specific behaviors in VR has been found to lead to more self-efficacy than video (Liu, Fan, Liu, & Ye, 2022) and traditional lecturing (Francis, Bernard, Nowak, Daniel, & Bernard, 2020). VR-based English learning has been deemed more self-efficacy conducive than traditional methods (Zheng, Young, Brewer, & Wagner, 2009). Research has qualitatively associated the gains in self-efficacy with the fact that VR affords hands-on experience and feedback for specific behaviors (Petersen, Klingenberg, Mayer, & Makransky, 2020). More broadly, virtual environment-based treatments, whether immersive or not, have been shown to lead to higher self-efficacy when compared to a control group that did not play and only completed a survey twice (Shu et al., 2019). However, it must be taken into account that the game treatments do not offer particular skills to be practiced and repeated towards mastery. Despite this difference, we hypothesize:

- H4.2: The game-based conditions will result in a larger environmental self-efficacy shift than reading the text.

We are also interested in observing relationships between immersion and self-efficacy. Previous research has provided qualitative evidence that behaviors are mimicked more easily with VR controllers

than with a mouse, which aids behavior memorization (Shu et al., 2019). Another study found that a feeling of presence was one of the mediators between usability and self-efficacy in a VR crane operating training system (Song, Kim, Kim, Ahn, & Kang, 2021). Avatar use has also been linked to increased nutrition efficacy via self-presence (Behm-Morawitz, Lewallen, & Choi, 2016). While these studies speak of the possible relationships between immersion/presence and self-efficacy in various areas, they tend to focus on the embodied practice of specific behaviors, which our design does not prioritize, or identification with an avatar separate from the player's identity, which is not the case either. While acknowledging these caveats, based on the evidence above we propose the hypothesis:

- H4.3: Self-reported immersion will be positively associated with a shift in environmental self-efficacy.

2.5. Exploring attitudes and self-efficacy in relation to enjoyment

In addition to the above hypotheses, we aim to explore the participants' self-reported climate change attitude and environmental self-efficacy shifts in relation to their interest/enjoyment. While links have been established between intrinsic motivation and pro-environmental behavior in daily life (Cooke, Fielding, & Louis, 2016; Pelletier, Tuson, Green-Demers, Noels, & Beaton, 1998), we seek to examine the specific space comprising enjoyment with a single, brief activity and reported changes in attitude as a result of participating. Given that attentional and perceptual biases exist in climate change engagement (Luo & Zhao, 2021), an enjoyable activity could reduce the person's predisposition to shape attention based on their pre-existing beliefs and motivations. In addition, a heightened motivation to play climate change games could influence motivation towards the climate through a recursive relationship (Vallerand, 1997). However, these possibilities remain only theoretical.

Regarding environmental self-efficacy, environmental amotivation due to a lack of competence has been correlated with negative feelings, and competence associated with motivation (Pelletier et al., 1999) and thus likely with positive affect. Interventions causing various emotions have been linked to pro-environmental action (Brosch, 2021), and positive affect in particular can be understood as a possible antecedent of climate change behavioral engagement when experienced or anticipated positive emotions are activated (Schneider, Zaval, & Markowitz, 2021). However, here we are interested in examining enjoyment from gameplay, rather than that from pro-environmental actions.

Thus, considering a lack of strong theoretical or empirical references apt to our intervention in the relationships signaled above, our first research question is:

- RQ1: How are the climate change attitudes and environmental self-efficacy data distributed in terms of interest/enjoyment?

2.6. Exploring pro-environmental intentions and behaviors

Finally, we are interested in engaging in a first exploration of the behavioral implications of our intervention. The relationship between pro-environmental intentions/behaviors and immersive/interactive experiences has been explored in the past (Ahn et al., 2014; Ahn, Fox, Dale, & Avant, 2015; Oh, Sudarshan, Jin, Nah, & Yu, 2020), generally in favor of the more immersive and interactive conditions. The role of attitudes and self-efficacy as predictors of PEI has been established in the literature, as well as the role of PEI in influencing PEB (Ajzen, 1991; Ajzen & Fishbein, 1980). In our intervention, all conditions involve a form of verbal commitment prior to the opportunity to engage in a particular PEB, which is a potentially equalizing factor as a predictor of PEB (Hines, Hungerford, & Tomera, 1987). Therefore, our second research question is:

- RQ2: What PEIs and PEBs can be observed as a result of the intervention?

3. Methods

3.1. Experiment design and participants

For this study, we conducted a lab experiment where participants were randomly allocated to three groups: a text-based stimulus (hereafter, control), a desktop screen-based 3D game (hereafter, PC), and an immersive VR version of the same game (hereafter, VR), while minding the gender distribution across groups as a potentially relevant variable for climate change engagement (Bloodhart & Swim, 2020; Gifford & Nilsson, 2014). The study follows the guidelines of the Finnish National Board on Research Integrity (TENK) and received the approval of Tampere University's Data Protection Officer.

Like previous literature involving the effects of immersive media, we are interested in medium-to-large effect sizes (see Breves & Schramm, 2021); that is, effect sizes approximately equal to or larger than Cohen's d = 0.5, or a difference of half a standard deviation (Cohen, 1988). According to a power analysis performed in Jamovi jpower, a one-tailed paired samples t-test would require a total sample size of 27 to reliably (with a power of 0.80) detect a minimum effect size of 0.5 with a maximum alpha of 0.05, which our sample (N = 105, with 35 participants per group) exceeds. Our sample size was constrained by the financial, time, and human resources available as well, so we aimed to recruit the largest possible sample given the desired statistical power.

The final sample was N = 105 participants (control n = 35; PC n = 35, VR n = 35). The average age (control = 29.29; PC = 28.57; VR = 30.63) and gender distribution (control: Female = 20, Male = 14, Other = 1; PC: F = 20, M = 14, O = 1; VR: F = 21, M = 13, O = 1) were similar in all three conditions. Most participants had attained some degree of university education (university of applied sciences bachelor's degree (n = 14), university of applied sciences master's degree (n = 2), university bachelor's degree (n = 27), and university master's degree (n = 48)), while a minority had primary education (n = 1), vocational school or course (n = 2), general upper secondary education (matriculation examination) (n = 10), or vocational college (post-secondary) (n = 1) education. The participants' origins included Finland (n = 35), Asia (n = 24), the EU (n = 23), the US or Canada (n = 5), non-EU Europe (n = 4), Africa (n = 3), Middle East (n = 3), and Latin America (n = 2). Six participants came from other areas, excluding Oceania.

3.2. Materials

3.2.1. Stimuli

This study uses a multi-platform (PC and VR) game designed and developed by the first author using the Unity game engine (version 2020.3.19f1)—*Climate Connected: Outbreak*. The game is single-player and story-based, meaning that the content is presented to all players in a similar order. The game frames climate change as a wellbeing issue by connecting its causes, physical mechanisms, and consequences to non-human and human life, with a focus on infectious diseases and pandemics.

Climate Connected: Outbreak can be described as a serious game since it consists of a system aiming to elicit engagement to produce outcomes beyond entertainment, but it also seeks to directly gamify participant behavior after playing through an email with climate action ideas, which will be described in the next subsection. Every step of the design process considered that the game would be multi-platform and used in research, and thus both versions needed to be as comparable as possible. Once development using the game engine began, the VR version was the first one to be implemented, given that its affordances are less conventional and we aimed to use the capabilities of the hardware while not introducing methods of interaction that would be trivial to implement and experience on PC but inadequate for VR.

In the game, players engage in three main forms of interaction. First, they can select dialog options when they encounter text-based

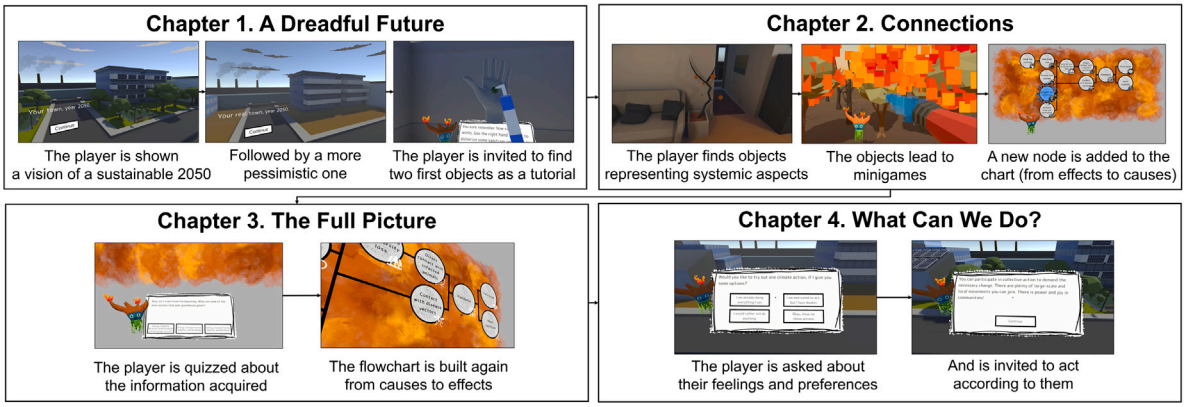


Fig. 1. The game's four chapters, including major events and actions within them.

information and prompts; therefore, they are able to progress through the explanations at their own pace, and occasionally select one of various choices leading to different feedback. Second, move through 3D spaces using a teleporting system. Third, select, grab, and throw objects in the space. The PC version is entirely mouse-based, where mouse movement rotates the camera and the left mouse button performs all actions described above. Meanwhile, the VR version requires pointing a controller at the element to be interacted with and pressing a trigger. While both systems are simple to use, VR permits a more physically involving interaction—where object manipulation is needed, VR players physically point their controller and hold the trigger to grab and need to move their arm and release the trigger to throw. The forms of interaction described for both systems should be rather natural to those less familiar with digital games or immersive VR, while the use of either controller in VR considers both right- and left-handed players. Finally, the choice of a teleporting-based locomotion system should minimize motion sickness.

The content is presented through four chapters. First, players find 14 items that represent elements of the climate-wellbeing system presented. The system is shown as a flowchart that players complete one node at a time—first finding a relevant object, then playing an associated minigame, and finally seeing how the new concept fits within the structure. The minigames are described, in order and with their associated node, in Table 1. After these, players complete a quiz reviewing the content and are invited to act for the climate if they wish to do so. For an overview of the game's structure, see Fig. 1. A video with more information has also been provided as supplementary content.

As elements potentially relevant to attitude, self-efficacy and behavior, the game includes (a) a guiding character that provides information and encourages players to explore the game and connect the game's insights to their own life and experiences; (b) textual, visual and interactive depictions of climate change-relevant aspects centering the player and the fictional local space, presented following climate change communication best practices (Sheppard, 2012; Wibeck, 2014); (c) motivational support adapted to the player's self-reported views (Roser-Renouf, Stenhouse, Rolfe-Redding, Maibach, & Leiserowitz, 2015) and based on self-determination theory (Cooke et al., 2016) and environmental amotivation-countering principles (Pelletier et al., 1999); and (d) proposed action profiles to act in the real world, thus going beyond consumer behavior (Stern, 2000; Wibeck, 2014). The behavior options are described at the end of Section 3.2.2.

As a control condition, participants were presented with a document that includes the same textual information, a series of flowcharts which also feature in the game, and narrative descriptions of the content encountered in the game.

3.2.2. Measurement instruments

Existing questionnaires were used to measure the participants' level of interest and enjoyment in the experience, as well as their immersion level, after the treatment. We also used existing instruments to measure climate change attitude and self-efficacy before and after the treatment. Additional PEI and PEB data were sourced from custom questionnaires.

Interest and enjoyment in the experience were measured via the Intrinsic Motivation Inventory's Interest/Enjoyment subscale (Center for Self-Determination Theory, n.d.) (hereafter ENJOYMENT), which consists of seven items presented in random order and rated on a 7-point Likert scale. The Cronbach's alpha value for the ENJOYMENT responses was 0.906, which denotes an excellent internal consistency. The data were computed as an average per participant.

Immersion was measured using the IMMERSION subscale in the GAMEFULQUEST instrument (Högberg, Hamari, & Wästlund, 2019), which aims to measure immersion in line with the interpretation presented in the background (Cairns et al., 2014). The Cronbach's alpha value for the IMMERSION responses was 0.714, which denotes an acceptable internal consistency. The data were computed as an average per participant.

To measure the participants' climate change attitude, we used the Climate Change Attitude Survey (CCAS), which was created to measure middle school "students' beliefs and intentions toward the environment with a focus on climate change" (Christensen & Knezek, 2015, p. 773) but has since been used with other samples, such as pre-service teachers (Oladipo, Awofala, & Osokoya, 2020). As previous studies have only arrived at tentative factors (Christensen & Knezek, 2015; Oladipo et al., 2020), we performed an exploratory factor analysis (EFA) on the pre-intervention participant responses using Jamovi 2.3.21.0 (package jmv 2.3.1). Bartlett's Test of Sphericity was adequate (<.001), and KMO measures of sampling adequacy showed most items to be adequate (0.8–1.0) and three to be middling (0.7–0.79), including Q10 (0.773), Q11 (0.792), and Q15 (0.775) (Kaiser, 1974).

The method involved minimum residuals and oblimin rotation, since it is a non-orthogonal option adequate for the assumption that the two subscales are correlated, as is the case. Parallel analysis detected a best fit with three factors. However, model fit measures reported unacceptable values (RMSEA 0.094, TLI 0.850) according to the literature (Hu & Bentler, 1999). Although it has been recommended not to use model fit indices in EFA due to their excessive sensitivity (Montoya & Edwards, 2021), the use of cutoff values questioned (Xia & Yang, 2019), and best practices suggesting that parallel analysis provides the largest number of plausible factors (Watkins, 2018), we first explored larger model configurations under the assumption that smaller RMSEA and larger TLI values indicate better fit (Xia & Yang, 2019). While models with 4 and 5 factors offered better fit according to these measures,

Table 1
The minigames in the order in which players encounter them; that is, from consequences to causes of climate change. Each minigame has an associated node, which are organized here according to more abstract categories. Interactions in the minigames are complemented with text explanations related to the issue being represented.

Category	Node	Minigame
Pandemic consequences	Face mask	The player is given a hand sanitizer bottle, which they must spray and spread on their hands. This minigame is part of the tutorial-like introductory chapter.
	Hand hygiene	The player is given a box of masks and they must throw them o approaching figures to prevent contagion. This minigame is part of the introductory chapter.
Climate change impacts on life	Direct contact with infected animals	The player embodies a bird pursued by a red cloud representing environmental degradation. Every time the player moves, the cloud extends. As the cloud spreads, the only option left is to migrate to the city.
	Contact with disease vectors	The player must place a mosquito net over a door before the time limit, applying glue first and then affixing the net to the frame. If they fail, the mosquitoes arrive with the nighttime and they must repeat the process.
	Habitat and biodiversity loss	The player is shown a forested area crossed by a river, and encouraged to build on it. As they select each segment, the forest gets cleared for construction and a river mammal's death is revealed as a consequence.
Physical manifestations of climate change	Droughts and wildfires	The player is surrounded by trees on fire which they can extinguish using a water hose that they have equipped, but no matter how fast they stop the fire, the trees cannot be saved.
	Extreme heat and heatwaves	The player is in the street during a scorching heatwave, represented by an orange hue covering everything and dead birds at different spots. The player must find the only safe place, a garage with air conditioning.
	Floods and storms	The player is in the middle of a flooded area. Since the stagnant water, reeds and accumulated leaves have allowed mosquitoes to breed in various places, they must find and remove such places.
	Land ice melting & Sea level rise	At the seafront of a town, facing the ocean and some distant ice formations, the player witnesses how sea level has risen in the past century and is forecasted to rise in the coming century, including the possibility of rise that submerges part of the surrounding town and surfaces multiple dead fishes around them due to warming.
Causes of climate change	Buildings	The player turns on water heating to prepare a hot bath, but this makes a fossil-based power plant appear behind as the infrastructure heating the water. They also turn on an air conditioning unit, which is revealed to emit potent greenhouse gases. The player turns them off before ending the minigame.
	Land use	The player is in a farm. With a water hose, they water some cereal crops, but when they have grown they are eaten by a cow, signifying the extensive use of cereals to feed animals. Then, as climate change progresses, the yield becomes smaller, but the animal keeps eating it. In the end, the player closes the barn doors representing the reduction of cattle farming to promote food security.
	Industry	The player is tasked with packing plastic toys. After this, a more pro-environmental option is presented, a toy made of wood, combined with the reduction in consumption.
	Energy	First, the player plays a shooting game in which each shot increases their "CO2 score", pointing towards the relationship between electronics and energy use. After this, they must disconnect the power going from a fossil fuel power plant and switch the energy source to wind turbines instead.
	Transport	The player sees the number of airplane tickets and soft drinks bought per second. Then, they can throw a hammer at the screen to represent a break with consumerism associated with, e.g., certain types of tourism.

we did not consider them to offer stronger theoretical explanations, as the variables seemed to refer to either (a) beliefs in climate change and its severity, or (b) attitudes regarding climate action, which are similar to the instrument's original theoretical foundations being beliefs and intentions (Christensen & Knezek, 2015). Thus, these models were discarded.

In the opposite direction, parallel analysis suggested one factor that was loaded by only two variables, which is a symptom of overfactoring (Gorsuch, 1983, cited in Watkins, 2018), with a third variable saliently loading (i.e., with a pattern coefficient ≥.3) on two different factors, which is also indicative of a sub-optimal solution (Watkins, 2018). We also used a scree plot as a subjective adjunct to the parallel analysis estimate (Watkins, 2018), which seemed to indicate a best fit with two factors, after which the magnitude change of the component eigenvalues was markedly reduced.

Given these procedures, which problematized solutions with three or more factors while pointing at two different theoretically justifiable constructs, we present here the two-factor model. Whereas factor 1 (climate change beliefs, CCBEL) seems to involve beliefs that climate change is real, concerning, and something that should be known and acted upon, factor 2 (attitudes towards individual climate action, CACT) focuses on attitudes towards the actions of individuals (often presented as I/we statements), which may also point towards intention to act as a component of attitude (Schultz et al., 2005) (see Table 2).

The Cronbach's alpha values were satisfactory for all six datasets involved in the analyses (CCAS pre = 0.868, post = 0.861; CCBEL pre = 0.869, post = 0.886; CACT pre = 0.781, post = 0.729). The values for the overall CCAS scale, CCBEL and CACT were computed both as average values for each participant before and after the intervention, and as a single score reflecting change between pre and post (post–pre).

To measure the participants' environmental self-efficacy (ESE), we used the 10-item version of the Environmental Self-Efficacy Scale (ESE-10) (Moeller & Stahlmann, 2019). The instrument asks respondents to rate a series of pro-environmental behaviors on a scale of zero to ten based on how certain they are that they are capable of doing them regularly. The Cronbach's alpha values were satisfactory for the pre (0.914) and post (0.880) values. The results were computed as averages and as a score reflecting change between pre and post.

To measure willingness to engage in climate action, participants were asked whether they wanted to select an area of climate action of interest to receive an email with a series of actions that they could do. Participants in the PC and VR conditions encountered this at the end

Table 2
EFA results on the pre-treatment CCAS scale.

Item	Descriptive statistics				Factors		h^2
	Mean	SD	Skewness	Kurtosis	CCBEL	CACT	
1. I believe our climate is changing	4.76	0.53	−2.58	7.85	**0.77**	−0.13	0.54
2. I am concerned about global climate change	4.55	0.73	−1.90	3.69	**0.76**	0.03	0.59
3. I believe there is evidence of global climate change	4.78	0.48	−2.14	3.95	**0.88**	−0.04	0.76
4. Global climate change will impact our environment in the next 10 years	4.65	0.71	−2.54	7.84	**0.73**	0.08	0.58
5. Global climate change will impact future generations	4.87	0.39	−3.10	9.59	**0.70**	−0.01	0.49
6. The actions of individuals can make a positive difference in global climate change	4.13	0.99	−1.06	0.39	0.19	**0.65**	0.55
7. Human activities cause global climate change	4.69	0.52	−1.41	1.07	**0.67**	0.10	0.51
8. Climate change has a negative effect on our lives	4.54	0.81	−1.97	3.98	**0.65**	0.03	0.44
9. We cannot do anything to stop global climate change (R)	4.29	0.91	−1.47	2.33	0.20	**0.45**	0.30
10. I can do my part to make the world a better place for future generations	4.25	0.72	−1.20	3.4	0.26	**0.36**	0.26
11. Knowing about environmental problems and issues is important to me	4.31	0.80	−1.44	2.84	**0.34**	0.23	0.23
12. I think most of the concerns about environmental problems have been exaggerated (R)	4.46	0.75	−1.40	1.74	**0.52**	−0.02	0.26
13. Things I do have no effect on the quality of the environment (R)	3.87	0.91	−0.59	0.04	0.11	**0.61**	0.43
14. It is a waste of time to work to solve environmental problems (R)	4.7	0.61	−2.90	12.7	**0.45**	0.13	0.27
15. There is not much I can do that will help solve environmental problems (R)	3.83	1.07	−0.80	−0.03	−0.12	**0.90**	0.74

Note. R = reversed item. h^2 = communality. Salient pattern coefficients ≥.3 in boldface.

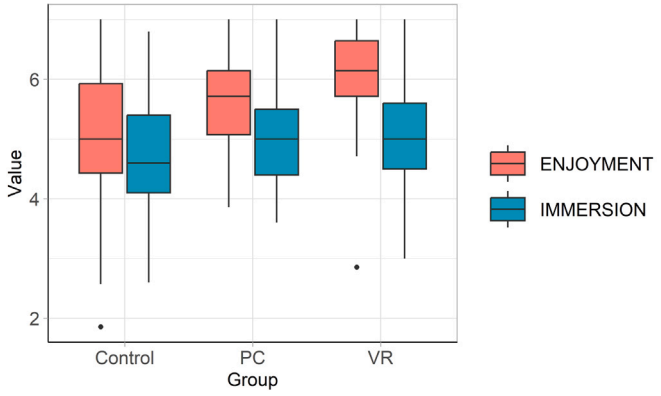


Fig. 2. ENJOYMENT and IMMERSION scores among the three participant groups.

of the game, whereas the control included this as an optional questionnaire after reading the text. The six options include advocate (be vocal and spread awareness), artist (create based on knowledge, feelings and perceptions of climate change), conscious citizen (do low-commitment political action), lone rider (take individual action), scholar (get to know more), and team player (support or engage in collective action), plus a refusal to engage. The participants' choice was recorded and they received an email with suggestions in accordance with it. For game players, the email was written from the perspective of the game's guiding character, and included a badge-like image reflecting their choice; text readers received only a text email written from the perspective of the research team. Finally, ten days after participating, they received a final survey with questions about their chosen action, including whether it had been done, and if it had inspired further action. The final email was also written from a different perspective depending on the participant's assigned condition.

3.3. Procedure

The study was advertised as an "experiment about immersion in climate change information" and compensated with one cinema ticket for participating. The experiment communication materials included a link to the initial survey, where prospective participants gave their informed consent and answered the pre-treatment questionnaire. Then, they booked a time for the experiment. They were told in advance to not wear glasses if possible, since VR may be used. Once they arrived at the experiment site, located in Tampere University, they were briefed on the process and invited to engage in a cognitive task related to climate change. Next, they completed a knowledge test and started the treatment itself. The participants were given the goal of getting "as good an understanding as possible of the content of the game/text". Both the control and PC conditions took place in the same computers where the test had been answered, located in individual cubicles with no outside view and soundproofing. The VR condition took place in a separate area that provided enough space for participants to move and employed an Oculus Quest 2 headset with two standard controllers. Participants were given a time frame of 30 to 60 min to finish the game or text and were informed that if they reached 60 min they would be invited to finish as soon as possible. Text readers who finished earlier than 30 min were allowed to re-read freely until the minimum time had been reached. After the treatment was completed, the participants immediately answered the post-treatment survey before moving on to other tasks, which included a second part to their cognitive exercise and knowledge test, and an optional interview.

4. Results

4.1. Enjoyment

First, we aimed to compare between the three groups to see whether playing the game led to higher ENJOYMENT than reading the text (H1.1), and whether immersive VR led to higher ENJOYMENT than PC (H1.2). Tests of normality (Shapiro–Wilk) and homogeneity of variance (Levene's test) indicated that the data does not meet the required assumptions for parametric testing. Therefore, a Kruskal–Wallis test was performed. The test found significant differences between the groups, $H(2) = 17.40$, $p < .001$. Post-hoc Dwass–Steel–Crichtlow–Fligner pairwise comparisons were performed and Cliff's Delta Calculator used for effect sizes. The tests revealed that VR users reported significantly higher enjoyment than PC users ($W = 3.66$, $p = 0.026$, $\delta = 0.359$) and text readers ($W = 5.60$, $p < .001$, $\delta = 0.549$), suggesting a medium and high effect, respectively (IIPUS, n.d.). The comparison between PC players and text readers was non-significant, $W = 2.72$, $p = 0.132$. Fig. 2 shows the ENJOYMENT values across the three participant groups.

4.2. Immersion

Next, we compared the three groups to examine whether technological immersiveness is correlated with IMMERSION (H2). Tests of normality and homogeneity of variance indicated that the data follows the required assumptions for parametric testing. Therefore, a one-way ANOVA was performed. Although the mean value was higher in the game groups (PC = 5.03, VR = 5) than in the control (4.76), the test did not find a significant difference between the three groups, $F(2, 67.5) = 0.848$, $p = 0.433$. Fig. 2 shows the IMMERSION values across the three participant groups.

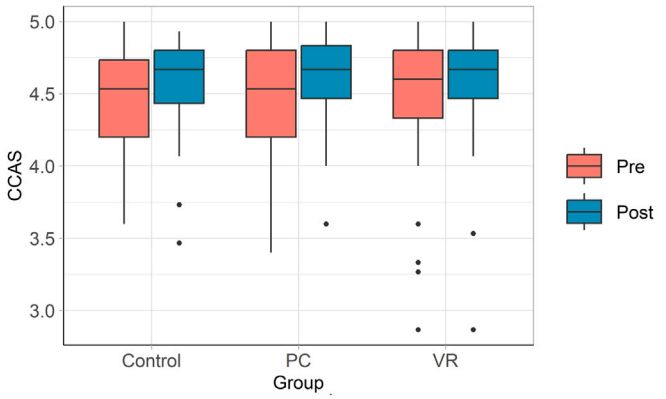


Fig. 3. Participants' CCAS before and after the treatment.

Table 3
Pre- and post-treatment comparison of participants' CCAS, CCBEL and CACT.

Group	Mdn		Z	p	r
	Pre	Post			
Climate change attitudes (CCAS)					
Control	4.53	4.67	85	p = .002	0.609
PC	4.53	4.67	87.5	p < .001	0.647
VR	4.60	4.67	69	p = .003	0.607
Climate change beliefs (CCBEL)					
Control	4.70	4.80	94.5	p = .011	0.500
PC	4.70	4.90	82	p = .015	0.495
VR	4.90	4.90	62.5	p = .011	0.547
Individual climate action beliefs (CACT)					
Control	4.20	4.40	91	p = .005	0.552
PC	4.20	4.20	61	p = .003	0.625
VR	4.00	4.20	99	p = .026	0.436

Note. Z = test value. p = significance. r = Rank biserial correlation.

4.3. Climate change attitude

4.3.1. Before–after differences

After focusing on ENJOYMENT and IMMERSION, we proceeded to examine whether the intervention had an effect on participants' climate change attitudes (H3.1), for which we compare the data before and after the intervention (see Fig. 3). Due to the data not meeting normality assumptions, we performed Wilcoxon signed-rank tests (one-tailed) for CCAS and the CCBEL and CACT constructs. The tests for each group showed a significant difference in the pre-post comparison in all three cases, as shown in Table 3. A Holm–Bonferroni correction was applied to each pair of results involving CCAS and either of its dimensions, CCBEL or CACT, for the same treatment group. All results remained significant at the 0.05 alpha level after applying the correction.

4.3.2. Comparison of attitudinal shifts between groups

Next, we examined possible differences in attitudinal effects between the game-based conditions and the control (H3.2) and between the three groups, where VR is expected to lead to a larger shift than PC and PC to a larger shift than the control (H3.3). For this, we performed an analysis of variance on the participants' attitudinal shift (that is, post–pre). Not meeting the assumptions for a parametric test, we performed a Kruskal–Wallis test, which found no significant differences in shift for CCAS ($H(2) = 0.209$, $p = 0.901$), CCBEL ($H(2) = 0.041$, $p = 0.980$) or CACT ($H(2) = 0.762$, $p = 0.683$).

To test whether IMMERSION was positively associated with shifts in climate change attitudes (H3.4), we first visually explored the data, which appears to be disperse rather than showing any discernible consistent patterns, as shown in Fig. 4.

Table 4
Spearman correlation tests involving IMMERSION and CCAS, CCBEL, and CACT.

		CCAS	CCBEL	CACT
IMMERSION	ρ	−0.003	0.039	−0.112
(all)	p-value	0.513	0.345	0.871
IMMERSION	ρ	−0.017	−0.074	−0.059
(control)	p-value	0.538	0.663	0.631
IMMERSION	ρ	−0.018	0.042	−0.172
(PC)	p-value	0.542	0.405	0.839
IMMERSION	ρ	−0.026	0.174	−0.123
(VR)	p-value	0.560	0.159	0.759

Note. Hypothesis is positive correlation.

Table 5
Pre- and post-treatment comparison of participants' ESE.

Group	Mdn		Z	p	r
	Pre	Post			
Control	6.40	7.20	150.5	p = .010	0.463
PC	6.30	7.60	81	p < .001	0.728
VR	6.70	7.50	161.5	p = .010	0.457

Note. Z = test value. p = significance. r = Rank biserial correlation.

In addition to the exploratory visualizations, a correlation matrix was created in Jamovi (see Table 4) to test H3.4. Given that visual examination indicated no clear linear relationships and normality tests revealed most of the data to be non-normally distributed, we decided against using the Pearson correlation coefficient. Instead, Spearman's rank correlation coefficient was used to assess the presence of a monotonic relationship between each pair of datasets. The results imply that IMMERSION is not significantly correlated with CCAS, CCBEL, nor CACT.

4.4. Environmental self-efficacy

4.4.1. Before–after differences

Next, we compared the participants' ESE data from before and after the intervention to see whether a change had occurred (H4.1) (see Fig. 5). Due to the data not meeting normality assumptions, we performed Wilcoxon signed-rank tests (one-tailed). The tests showed a significant difference in the pre-post comparison (see Table 5).

4.4.2. Comparison of ESE shifts between groups

Next, we examined possible differences in ESE effects between the game-based conditions and the control (H4.2). For this, we performed an analysis of variance on the pre-post change in ESE to examine possible differences between groups. Not meeting the assumptions for a parametric test, we performed a Kruskal–Wallis test, which found no significant differences in ESE shift ($H(2) = 4.06$, $p = 0.131$).

To test whether IMMERSION was positively associated with a shift in ESE (H4.3) we first visually explored IMMERSION plotted against shifts in ESE (post–pre). The data does not seem to show any discernible consistent patterns, as shown in Fig. 6.

In addition to the exploratory visualizations above, a correlation matrix was created in Jamovi (see Table 6) to test H4.3. Due to no clear linear relationships and non-normal distribution, Spearman's rank correlation coefficient was used to assess the presence of a monotonic relationship between each pair of datasets. The results imply that IMMERSION is not significantly correlated with ESE.

4.5. Exploration of relationships between interest/enjoyment and climate change attitude and environmental self-efficacy

To answer RQ1, which aims to explore the distribution of attitudes and self-efficacy in relation to interest/enjoyment, we conducted data visualization-based explorations. The visualizations involving ENJOYMENT plotted against shifts in CCAS and its factors or ESE (post–pre) did not seem to yield any discernible consistent patterns, as shown in Fig. 7.

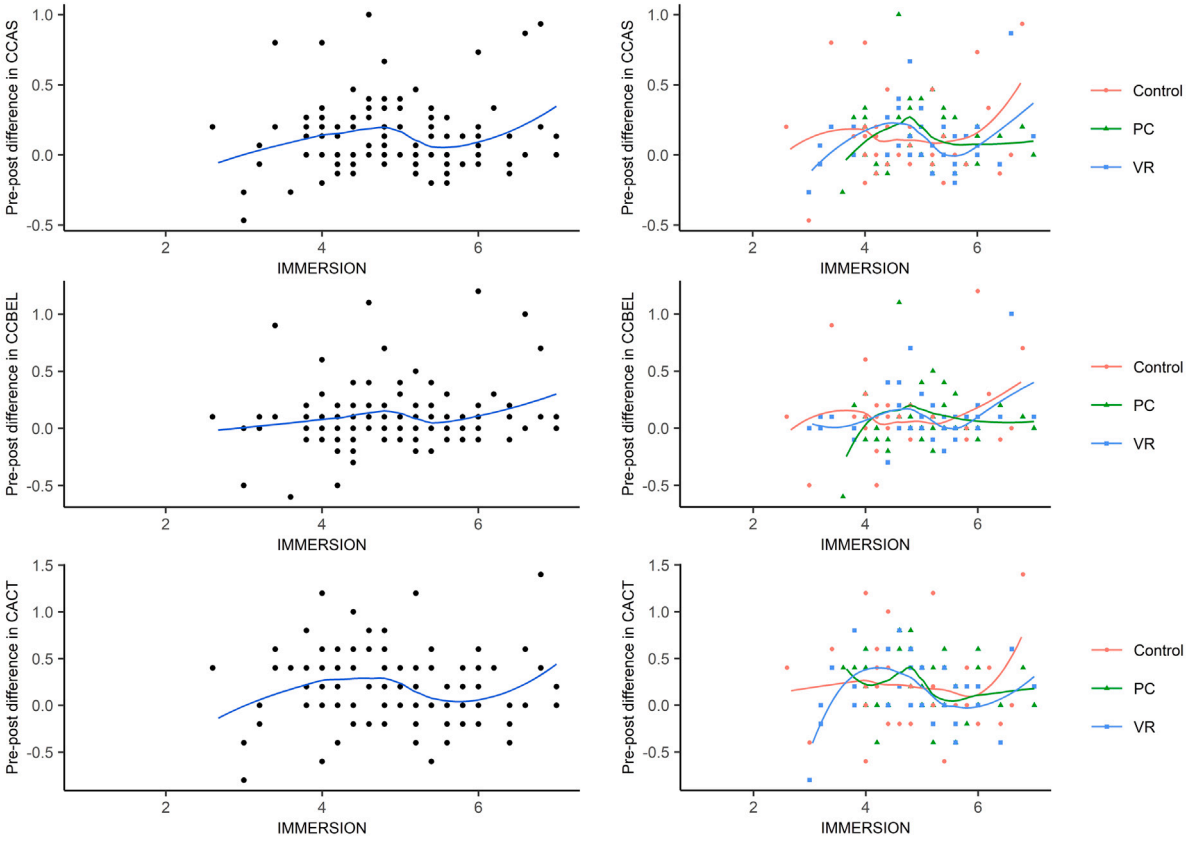


Fig. 4. Shift in CCAS, CCBEL and CACT based on IMMERSION for all participants (left) and divided by treatment group (right).

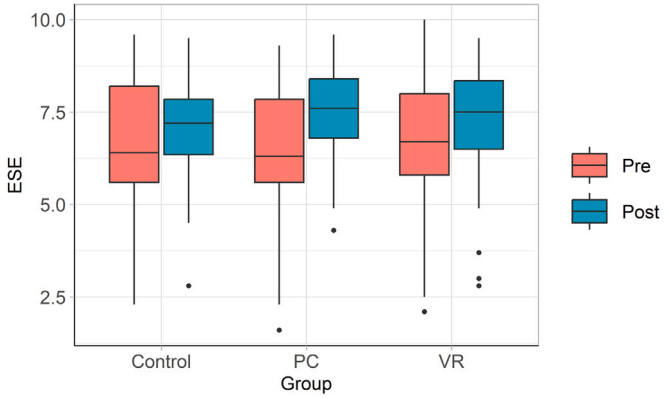


Fig. 5. Participants' ESE before and after the treatment.

Table 6
Spearman correlation tests involving IMMERSION and ESE.

		ESE
IMMERSION (all)	ρ	−0.093
	p-value	0.828
IMMERSION (control)	ρ	−0.204
	p-value	0.880
IMMERSION (PC)	ρ	−0.309
	p-value	0.964
IMMERSION (VR)	ρ	0.139
	p-value	0.212

Note. Hypothesis is positive correlation.

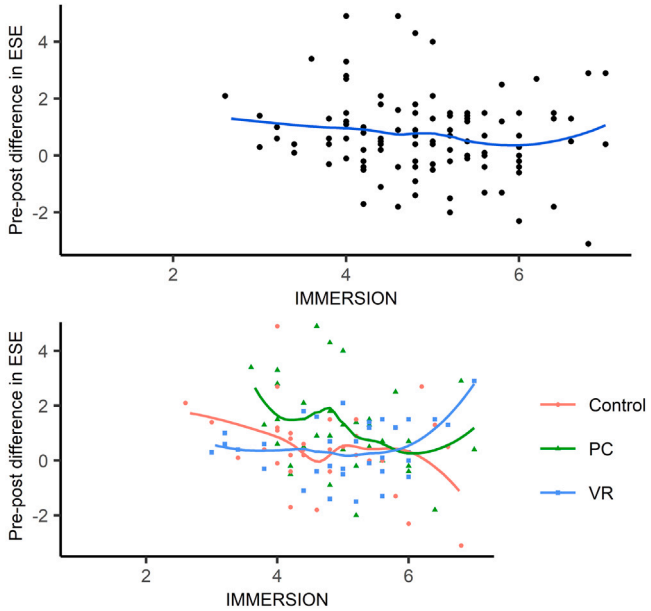


Fig. 6. Changes in ESE based on IMMERSION for all participants (top) and divided by treatment group (bottom).

4.6. Exploration of pro-environmental intentions and behaviors

Of 105 participants, 102 signaled their willingness to engage in climate action at the end of the game (35 in each group) or after reading the text (32). Because some of the expected values were under

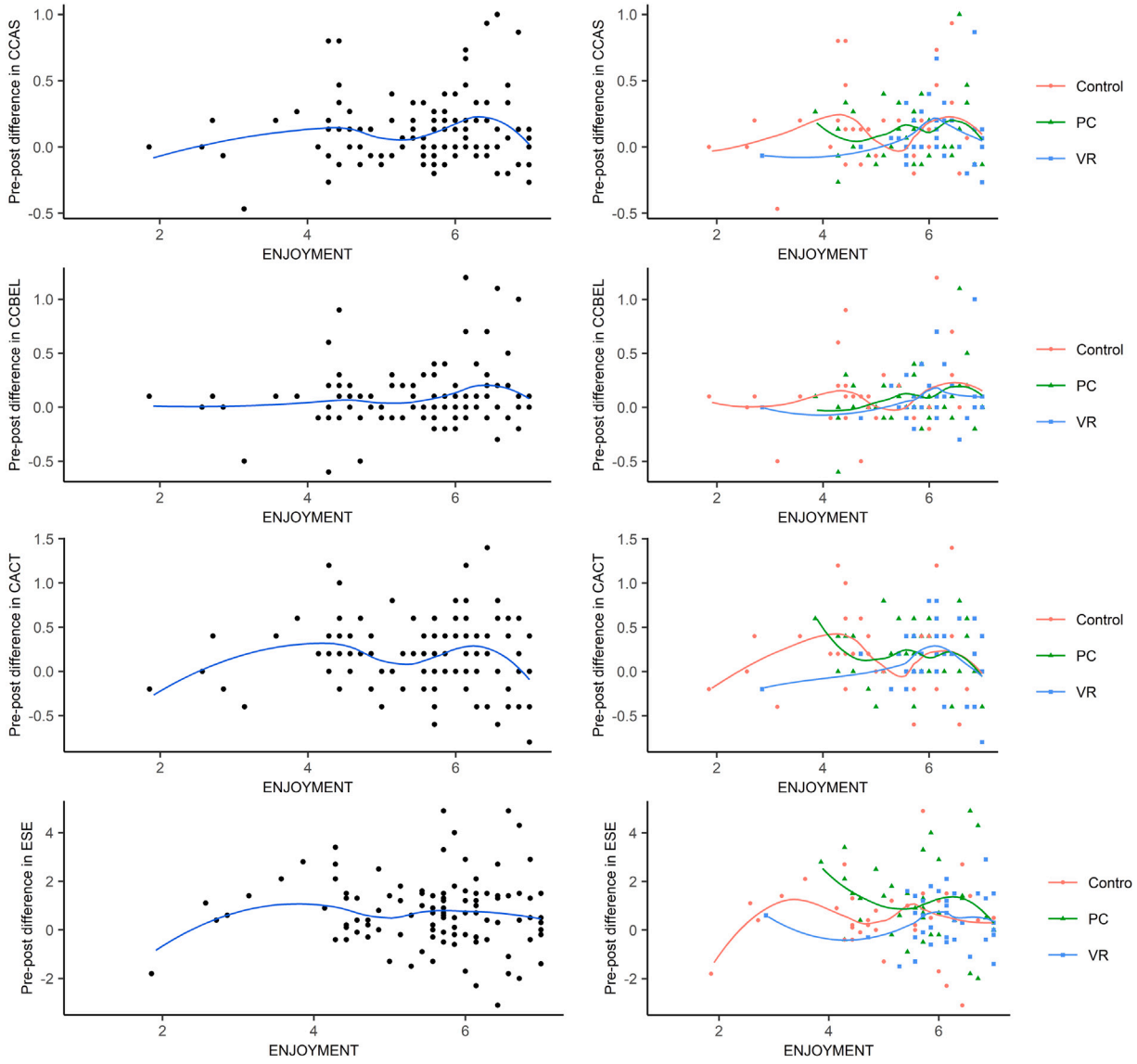


Fig. 7. Changes in CCAS, CCBEL, CACT and ESE based on self-reported ENJOYMENT for all participants (left) and divided by treatment group (right).

5, we conducted a Fisher's exact test (FET) to detect possible significant differences between the three groups. However, the result was non-significant ($p = .105$).

Of the 102 participants who chose to receive action proposals, 28 chose to receive further information to read, 25 selected individual actions, 21 chose collective action, 12 selected civic actions, 8 selected advocacy, and 8 chose to receive art-related suggestions.

Of the 102 participants, 42 answered the follow-up questionnaire sent 10 days later. Of these participants, 11 had chosen individual action, 9 information, 8 collective action, 7 civic action, 4 advocacy, and 3 art-related actions. Fifteen participants self-reported having done one climate action, 9 declared having taken steps towards it, 12 said that they would do it in the future, 3 said that they had not done their intended action, and 3 declared having forgotten their intended action.

The respondent distribution is similar when comparing by group (control: 14; PC: 13; VR: 15), as are their specific answers regarding action completion and their opinion on whether they had been inspired to take further action after participating in the experiment (see Table 7). The results of χ^2 tests of independence (or FETs, if any of the expected values was under 5) involving each of the six categories in Table 7 yielded non-significant results, indicating that there are no significant differences between the three treatment groups in any of them.

Table 7
Self-reported participant climate action completion after the experiment.

Group	Done	Steps	Future	No	Forgot	Inspired
Control	6	3	4	1	0	11
PC	3	4	3	1	2	11
VR	6	2	5	1	1	12
Total	15	9	12	3	3	34

5. Discussion

In this article, we have tested various hypotheses related to the use of a narrative climate change game in PC and VR and its comparison to a control, a text-based informational equivalent (see Table 8 for a summary of statistical testing outcomes). Our findings suggest that all three conditions increased climate change attitudes and environmental self-efficacy, but the study did not find significant differences between the three conditions. We found significant differences between the self-reported enjoyment of VR players and those in the other groups, which supports our hypothesis that playing in VR would be more enjoyable than playing on PC and partially supports our hypothesis that playing the game would be more enjoyable than reading the text. However, we

Table 8
Summary of hypotheses, including the statistical techniques used and the outcome.

Hypothesis		Statistical test	Outcome
H1.1	Playing the game in any form will be significantly more enjoyable than reading the text.	Kruskal–Wallis & Dwass–Steel–Crichtlow–Fligner	Partially supported
H1.2	Playing the game in VR will be significantly more enjoyable than playing on PC.	Kruskal–Wallis & Dwass–Steel–Crichtlow–Fligner	Supported
H2	Technological immersiveness will be positively associated with self-reported immersion.	One-way ANOVA	Not supported
H3.1	The intervention in any of its forms will positively affect participants' climate change attitude.	Wilcoxon signed-rank	Supported
H3.2	The game-based conditions will result in a larger climate change attitude shift than reading the text.	Kruskal–Wallis	Not supported
H3.3	Technological immersiveness will be positively associated with a shift in climate change attitude.	Kruskal–Wallis	Not supported
H3.4	Self-reported immersion will be positively associated with a shift in climate change attitude.	Spearman's rank correlation coefficient	Not supported
H4.1	The intervention in any of its forms will positively affect participants' environmental self-efficacy.	Wilcoxon signed-rank	Supported
H4.2	The game-based conditions will result in a larger environmental self-efficacy shift than reading the text.	Kruskal–Wallis	Not supported
H4.3	Self-reported immersion will be positively associated with a shift in environmental self-efficacy.	Spearman's rank correlation coefficient	Not supported

did not find significant differences in self-reported immersion across conditions, as well as significant correlations between immersion and changes in climate change attitudes and environmental self-efficacy. We also explored those variables in relation to enjoyment, finding no clear data patterns. Finally, our exploration of participant behavior yielded similar results for all three conditions, both in commitment to action and self-reported completion ten days after the intervention.

The findings regarding enjoyment were partially surprising, as it was expected that PC players would have found playing a game significantly more attractive than reading a text. However, there is precedent about games not being more motivational than traditional instruction in educational settings (Wouters, Van Nimwegen, Van Oostendorp, & Van Der Spek, 2013). In this intervention, the use of a text in a narrative style, with rich descriptions and direct references to the reader, and supported by flowcharts, may have contributed to a heightened interest if compared to a hypothetical text missing those features. At the same time, the nature of the experiment required the game conditions to be rather text-heavy, which may have reduced the distance between the different experiences. While immersive VR may be attractive due to its novelty and capacity for stimulation, interacting with a computer can be seen as a more mundane activity. Furthermore, the fact that the experiment was advertised as concerning climate change and the participants enrolled voluntarily may have attracted a sample that was particularly willing to engage with environmental content irrespective of the format. In addition, the experiment's setting, a university lab, may have also attracted a sample who is used to reading and is generally interested in text-based science dissemination.

The fact that no significant differences in self-reported immersion were found between conditions was also surprising. While the measurement instrument used, the "Immersion" subscale of GAMEFULQUEST, refers to immersion as a construct that signals involvement with the game rather than spatial presence, or the feeling of being transported to a different space (Cairns et al., 2014), it was nonetheless hypothesized that participants would report higher levels of immersion in the game conditions, and in VR in particular due to its higher potential for cognitive absorption (Kampling, 2018). Once again, the fact that the text was generally perceived as interesting and the game was largely text-based may have affected the audience's engrossment to an extent that no significant difference could be found, especially given the existence of narrative transportation as an immersive phenomenon (Green, 2021).

Participant attitudes were in general high before the stimulus, with a median value of 4.6 out of 5, which adds to the perception that this was a rather environmentally aware (and predisposed) sample. While the margin for increment was quite limited, the intervention increased the attitudes with a generally medium effect size. The fact that the action-related attitudinal component had a median significantly lower than the climate change beliefs component (4.2 vs. 4.8) implies that the general trust in one's own role as mitigating agents somewhat lagged behind interest and concern.

The findings on the intervention's attitudinal effects require further discussion, since three of the four proposed hypotheses were rejected. While we expected that all the conditions would have an impact on the participants' attitudes, the lack of difference between their effects may be explained by the similarities in the techniques employed. As both the text and the game largely relied on written communication to transmit information, the added value of playing the game (i.e., via features such as goal- and exploration-based interactive discovery of the story, a guiding character, graphical elements giving substance to the descriptions, and the possibility to choose dialog options) was not enough to drive significantly different attitudinal outcomes. As the use of a simulation in a similar experimental design also provided a similar result (Soekarjo et al., 2015), our study adds to the existing evidence with a narrative linear game and the involvement of an immersive VR condition. Regarding the role of VR, a comparison study showed VR visualizations raising environmental awareness as much as other methods, but not significantly more (Thoma et al., 2023). Where differences have existed, these tend to be attributed to a higher presence experienced in immersive VR, which was not observed in our study (Thoma et al., 2023). While we did not find a clear correlation between immersion and climate change attitudes, our results do not directly contradict others' assumption that presence can amplify media effects, especially given the different ways in which immersion and presence can be measured. Since it has been suggested that visual realism may not be a relevant aspect in shifting attitudes (Thoma et al., 2023), which supports the potential of cartoon-like environments like the ones used here, the effort to create experiences that foster presence in a clearer way may focus on creating more vivid stimuli (Bailey et al., 2015), that is, higher protagonism of the 3D multimedia environment and interaction with it over text-based communication.

The fact that there were no differences between the groups in terms of self-reported immersion also renders the lack of correlation with attitude somewhat unsurprising. The lack of an apparent relationship between enjoyment and attitudes/self-efficacy may also be explained partially through the fact that participants came to the experiment voluntarily, so even without experiencing a large degree of enjoyment, they may have generally paid attention to the text or game and attempted to do as best as they could.

In terms of self-efficacy, the similarity of the results may be explained as well by some of the arguments put forward when discussing attitude. While not comparatively as high, the participants' median self-efficacy before the intervention was rather considerable (6.6/10), although by the end it was raised to a median of 7.4. It is apparent as well that the additions provided by the game, both interactive and in terms of visualization, were not enough to drive different self-efficacy outcomes. This can be explained by the fact that the game did not feature situations where behavioral learning was practiced, but it rather supported self-efficacy through textual elements. Instead of simulating environmental challenges and providing opportunities for problem-solving and skill-building, most of the game used challenging situations to depict climate change issues, leaving the bulk of the solutions-oriented part for the end. Finally, when examining the three conditions individually, it can be seen that PC had a noticeably, although non-significant, larger effect than text and VR on environmental self-efficacy. Some possible explanations for its advantage over immersive VR include its potential ease of use and familiarity, as well as the fact that players in immersive VR may be more inclined to explore their surroundings while experiencing more difficulty and tiredness when reading (Knaack, Lache, Preikszas, Reinhold, & Teistler, 2019).

Our final area of focus in this article, the participants' PEI and PEB, yielded no differences in patterns when comparing the three conditions. This is consistent with the lack of difference in attitudes and self-efficacy. Even so, 34% of the experiment participants answered the final survey declaring that they had either done a climate action, progressed towards it, or planned to do it in the future. While it is possible that some of these actions would have been done even without the intervention, the result provides a point of comparison for similar future research.

5.1. Limitations and future research avenues

This study had some limitations that should be acknowledged, pointing at the same time towards future research avenues. First, and as said above, the design involved an experiment where willing participants were recruited. Although they received a small compensation for their participation, it can be assumed that a large proportion of them were particularly interested in the topic and were used to consuming information about it. Since our sample comprises participants that were generally concerned about climate change, young, and formally educated, our study should be complemented with other samples and in other contexts to arrive at a truly representative picture. Future research should also take into account that, despite the lack of clear comparative advantage in terms of attitudinal and self-efficacy improvement, games may be more attractive to audiences who may not be willing to read a text, as well as offer potential memorability (Pfirman et al., 2021) and recommendation advantages. Qualitative techniques such as interviews or focus groups may offer deeper insights into the participants' thoughts and opinions (Fernández Galeote et al., 2022).

The sample size is another limitation, since it may have been insufficient to detect small effects. The statistical examination of PEI and especially PEB will benefit from larger sample sizes in future studies, since in these cases the data were smaller than the overall sample. Therefore, the statistical test results referring to PEI and PEB reported here should be interpreted critically and need examination with larger samples.

Regarding the behavioral outcomes, two limitations should be taken into account. First, all of the groups included a prompt to signal the intention to participate in climate action. Therefore, we cannot isolate the effects of this commitment. This prompt was shown at the end of the game and immediately after reading the text, which could have influenced somehow the participants' responses to the subsequent survey by introducing a bias where the commitment impels participants to be consistent in their subsequent behavior, including reporting their attitudes (Lokhorst, Werner, Staats, van Dijk, & Gale, 2013). Second, we did not compare our participants to others who had not been exposed to a comparable stimulus that would offer a true baseline for pro-environmental behavior. Thus, this intervention should be complemented with others that test attitude change and propensity to act without offering an explicit commitment, nor a comparable treatment.

The measurement methods used also have limitations. First, in terms of what may lie beyond our capacity to observe and report: as long as attitude and self-efficacy depend partially on pre-existing personal variables, such as emotions, ideology, and physical and social constraints (Kollmuss & Agyeman, 2002), self-reported measures will remain difficult to assess and predict, including their possible translation into hope or productive engagement, as even the most engaged person needs times of disconnection (Pihkala, 2022). Second, we measured behavior through voluntary self-report ten days after the intervention. We can only assume that those who did not answer the survey did not do anything, and that those who answered were fair in their reporting. Other methods of assessing behavior may complement our findings.

A further limitation involves the statistical analyses performed. Due to there being different constructs of interest, multiple statistical tests have been done, which can raise the likelihood of detecting spurious relationships. However, the results found are generally clear in terms of significance, and corrections have been applied to those cases in which multiple tests referred to the same population and hypothesis and significant results were found.

Finally, we would like to echo previous works (Fernández Galeote et al., 2021; Soekarjo et al., 2015) in continuing to encourage rigorous designs where games and immersive media are compared to informationally similar conditions. Rather than assessing if games and/or immersive VR can impact climate change engagement positively, which seems to be a rather consistent finding, it is time to understand more deeply and in context how they do so, and what their comparative advantages may be.

CRediT authorship contribution statement

Daniel Fernández Galeote: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Visualization, Writing – original draft, Writing – review & editing. **Nikoletta-Zampeta Legaki:** Conceptualization, Methodology, Formal analysis, Funding acquisition, Investigation, Visualization, Supervision, Writing – review & editing. **Juho Hamari:** Conceptualization, Methodology, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.chb.2023.107930.

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